8D report

Customer plant	Eurotec	**Supplier**	AVO Carbon Germany	**Date of claim**	01.02.2024
Customer contact	A. Strnatkova	**AVO contact**	G. Luschtinetz		
Customer reference	24007	**AVO reference**	240109		

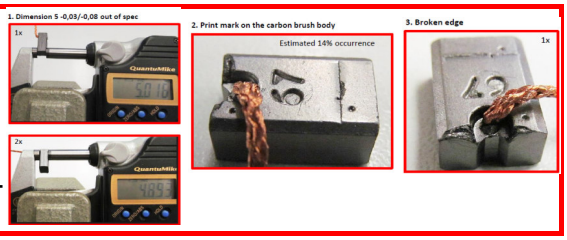
QUICK RESPONSE (D1-D3)
This document must be issued within 24 hours after claim notification

D 1. Description of the problem

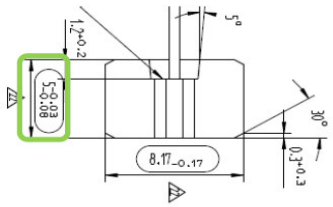
1.a What is the problem ? Use 5W+2H to describe what is the current situation.

Problem description	1.Dimension 5 -0.03/0,08 mm out of spec (1x measured 5,016mm, 2x measured 4.893 mm) 2.Print mark on the carbon brush body (estimated 14% occurrence) 3.Broken edge (1 pc)	What happened?	•Dimension 5 -0.03/0,08 mm out of spec (1x measured 5,016mm, 2x measured 4.893 mm) •Print mark on the carbon brush body (estimated 14% occurrence) •Broken edge (1 pc)
Incident type	Customer claim	Why is it a problem?	Parts can not be assembled.Out of spec.
PN customer/AVO	27579.52267 / 1020510	When did it happen?	No information
Part description	Carbon brush	Who detected it?	No information
AVO batch number	273526	Where was it detected?	No information
Recurrence	non reccurent	How was it detected?	No information
Defect S/R	no	How many nOK parts ?	9948
Standard process used	yes		

nok part picture / sketch



ok part picture / sketch



1.b Problem solving team

Name	Position	Location	Email
	QM	AVO Germany	quality-avo-Germany@avocarbon.com
Martin Röhr	PRD	AVO Germany	
Mustafa Yorulmaz	KVP	AVO Germany	
Joachim Gehl	QM	AVO Germany	
Giovanni Balenzano	PRD	AVO Germany	

1.c Tools to use

yes	**Ishikawa**		**5why's**		**LLC**		**Audit**		Other

1.d First analyse of problem / potential root causes

Potential root cause(s):

The samples are mandatory for the deeper analysis. Feel free to use our FedEx account. (Please see below marked in yellow)

Samples of nok parts required in plant for analyse: yes

Express shipping number account: FedEx 186141300

D 2. Risk on similar products and processes

Can this defect happen on similar parts ?

		Comments (compulsory if yes)
Other family product	---	
Other line	---	
Other tools/work station	---	
Other	---	

D 3. Containment

Sorting criterias (add additional sheets if needed)

Location in the supply chain	Nb of involved parts	Nb parts sorted	Nb nok parts	ppm	Date	Resp.
Parts at customer	**13.948**	---	**9.948**	---	01.02.2024	A. Strnatkova
Parts in transport		---	---	---	---	---
Parts at AVO Germany stock	**28.000**	---	---	---	---	---
Parts at AVO production	---	---	---	---	---	---

First contained shipment information: 28.000 pcs. need to be sorted before shipment with special labelling

Immediate actions to prevent production and delivery of nok parts ?

Actions	Resp.	Date plan.	Date clos.
Register Complaint	G. Luschtinetz	02.02.2024	02.02.2024
Information of production	G. Luschtinetz	02.02.2024	02.02.2024
Quality alert	G. Luschtinetz	02.02.2024	02.02.2024
Check the production-, quality- and maintenance data	G. Luschtinetz	02.02.2024	02.02.2024

8D report

PLAN - DO (D4-D6)

D 4. Root cause of occurrence

Work out the root causes using Ishikawa and 5 why's

Root cause of occurrence :

to 1: TRC:Based on the ground surface the grinding tool was worn and the lifetime cycle was not reached.
MRC: The lifetime cycle was wrongly defined.
to 2: The mark on the brush is caused by the part clamp. The angle was lower than 90° a high point in the end of the surface occure.During the input of the brush the high point scratch carbon brush. Due to the wrong insert angle of the brush the back side are also damaged.
to 3: The downholder spring was broken and during the brush was pushed through the transfer guiding rail, the brush was not positioned and pushed against the sharp edge of the welding gauge.

How was the defect reproduced ?

to 1: Not applicable
to 2: Used the part mounting.
to 3: Manipulation of the downholder spring of the welding gauge.

D 5. Root cause of non detection

Work out the root causes using Ishikawa and 5 why's

Root cause of non detection :

to 1: The tool was earlier worn than the defined lifetime cycle. Because of that the tool was not changed and the hourly sample inspection could not detect the too big parts.
to 2: The hourly did not detect the part.
to 3: Did not use the red box during setup

D 6. Corrective actions and non detection

Actions	Resp.	Date plan.	Date clos.
to 1:Reduce the lifetime cycle from 500.000 to 400.000	M Röhr	20.02.2024	
to 2: Hightlight scratches in the IQS-SPC Tool with a picture that the scratch section of the book of defect is not allowed on these parts	M. Altun	22.02.2024	
to 2: Change the manufacturing tolerance on only upper 90° angle is allowed to prevent high points in movement directions in the drawing	U. Wallocha	04.03.2024	
to 3: Sensiblization of the workers	J. Gehl	27.02.2024	
to 3: Add the spring to the preventive maintenance with monthly cycle	P. Balenzano	26.02.2024	
to 3: Re-design of the gauge with slighty chamfers to avoid bad adjustment	U. Wallocha	tbd	

CHECK - ACT (D7-D8)

D 7. Effectiveness of action plan

Actions to check effectiveness (to plan with part D6)	Resp.	Date plan.	Date clos.
to 1: Check next batch on print marks	Team	next production	
to 2: Workplace audit	Audit team	next production	
to 3: Workplace audit	Audit team	18.03.2024	

D 8. Lessons learned

8.1 Do following documents require an update ?

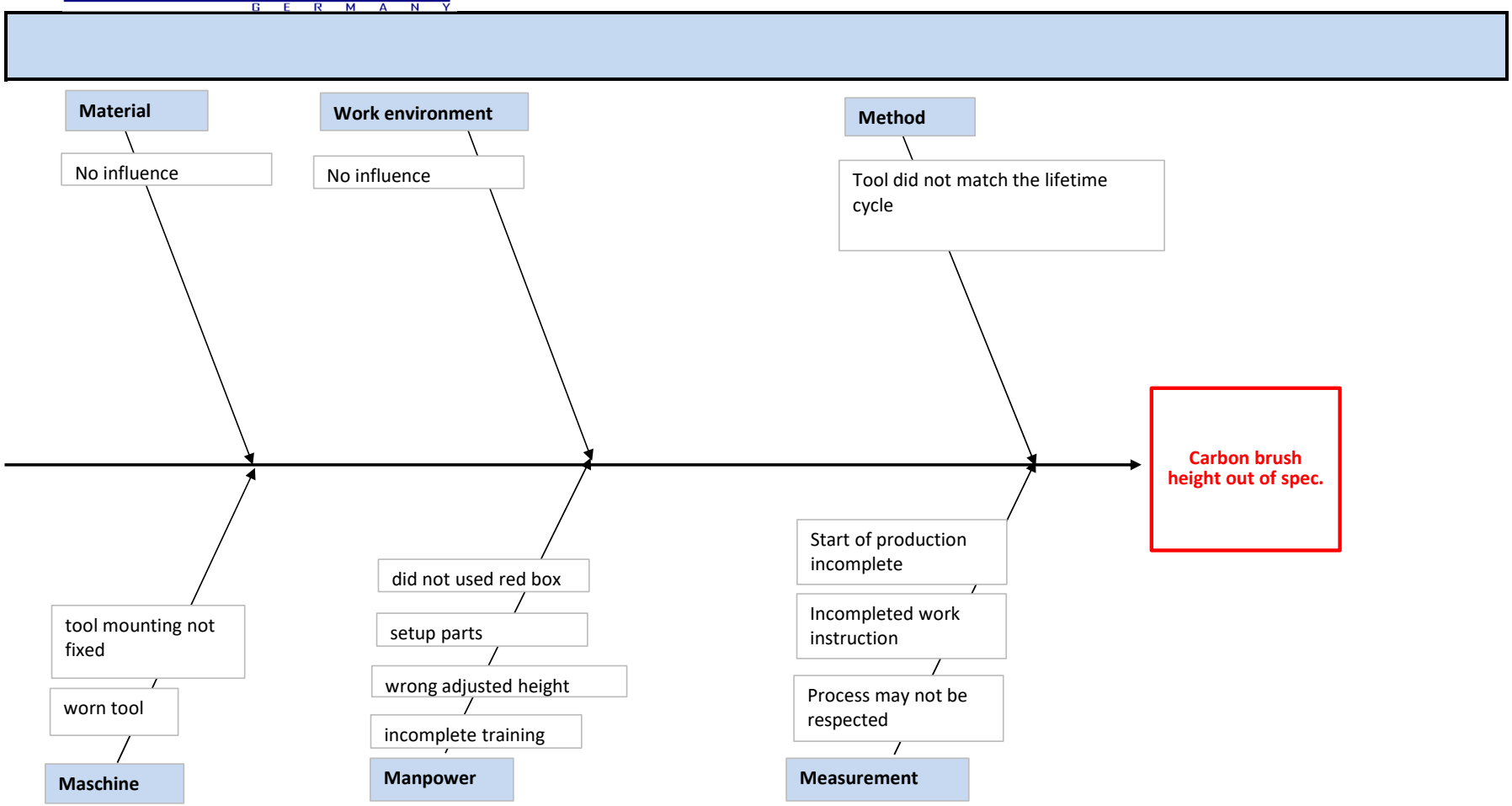
	Resp.	Date plan.	Date clos.
PFMEA	Mr. OTT		
		verified	changed
			yes / no
Control Plan	Mr. OTT		
		verified	changed
			yes / no
Work instruction			
Control instruction			
Procedure			
Control gage			
Maintenance instruction			
Lesson Learned Card			
Other			

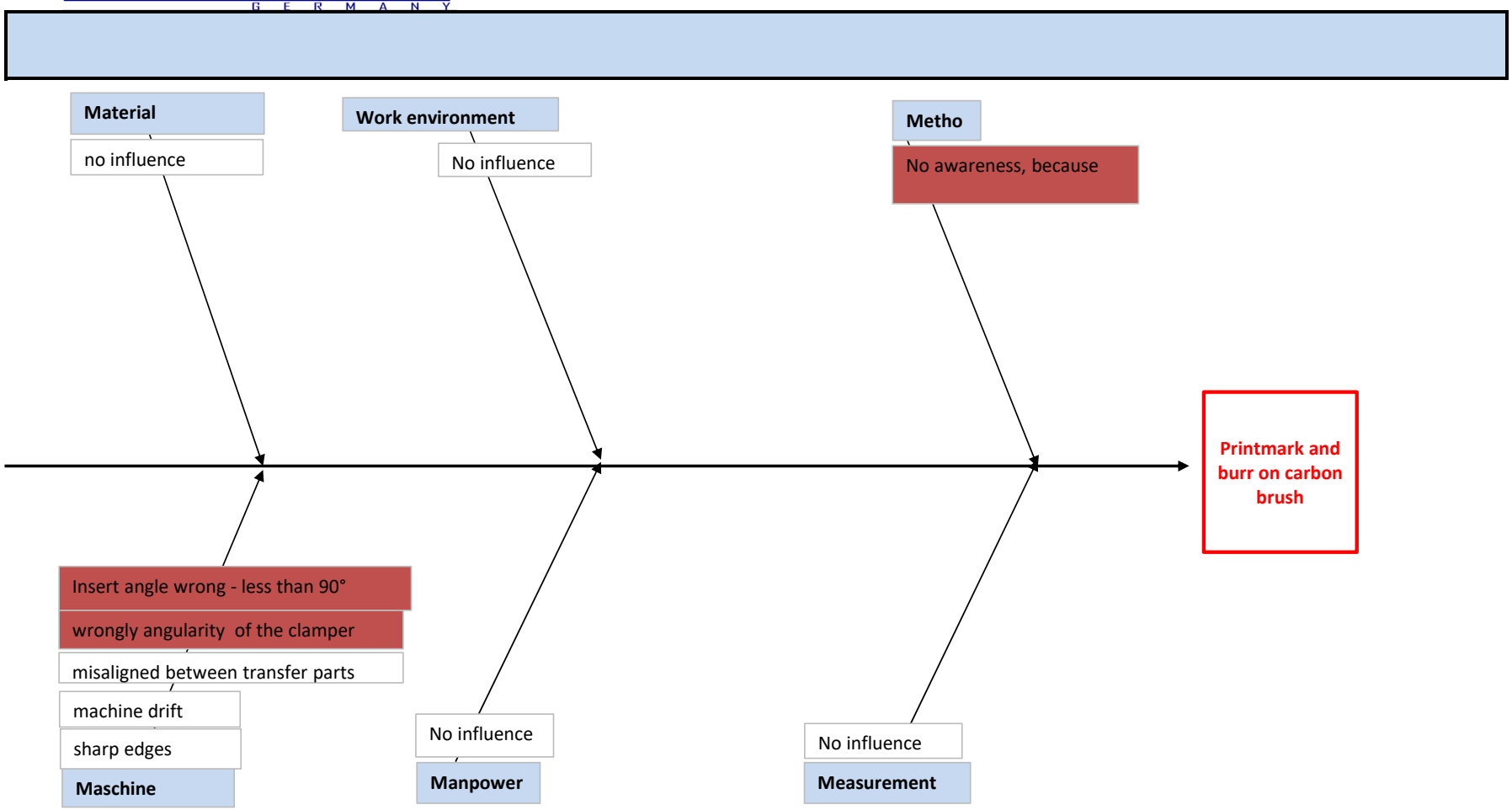
8.2 Generalization of countermeasures to similar lines/ products

8.3 Internal audit (if required)

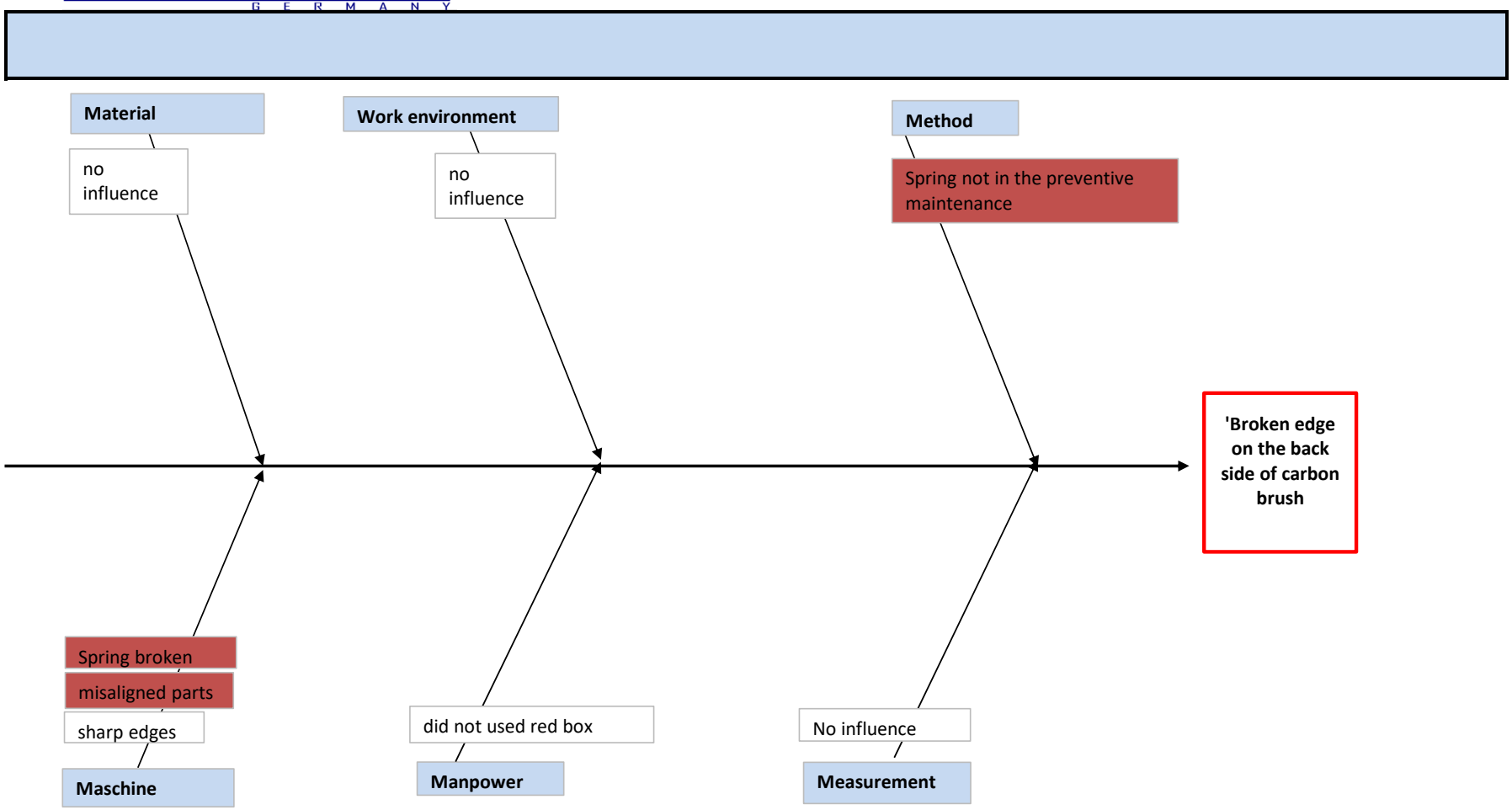
Date Results

Comments

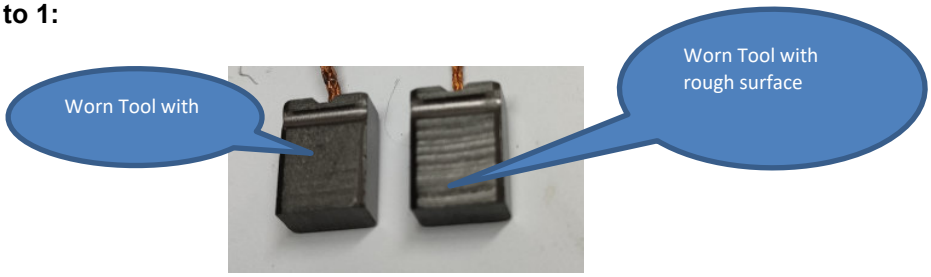

Material
No influence
Work environment
No influence
Method
Tool did not match the lifetime cycle
Carbon brush height out of spec.
tool mounting not fixed
worn tool
Maschine
did not used red box
setup parts
wrong adjusted height
incomplete training
Manpower
Start of production incomplete
Incompleted work instruction
Process may not be respected
Measurement


Material
no influence
Work environment
No influence
Metho
No awareness, because
Printmark and burr on carbon brush
Insert angle wrong - less than 90°
wrongly angularity of the clamper
misaligned between transfer parts
machine drift
sharp edges
Maschine
No influence
Manpower
No influence
Measurement

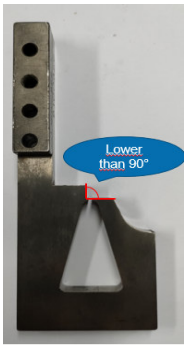
Ishikawa


Material
no influence
Work environment
no influence
Method
Spring not in the preventive maintenance
'Broken edge on the back side of carbon brush
Spring broken
misaligned parts
sharp edges
Maschine
did not used red box
Manpower
No influence
Measurement

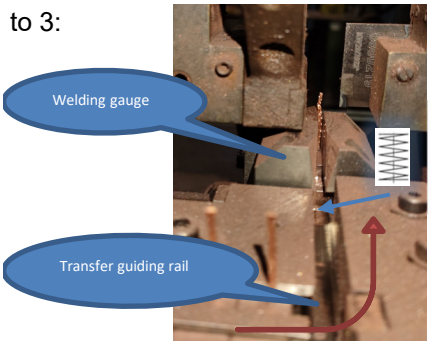
to 1:


Worn Tool with
Worn Tool with
rough surface

to 2:


Lower
than 90°

to 3:


Welding gauge
Transfer guiding rail